

CPR Memo

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1 Note about t

The date isn't continuous for the data, for example, let's assume date of row1 = 2020.12.23, and date of row 2 = 2020.12.26. Although there is 2 days between 2020.12.23 and 2020.12.26, but in the case we use, if $t=2020.12.23$, then $t+1 = 2020.12.26$

2 Exchange rate data collection and cleaning

1. We downloaded exchange rate data from Bloomberg using the code "@BDH". In total, we downloaded data for 23 currencies according to the CFETS baskets from the beginning of 2016 to the summer of 2024. For each currency, we download the **per USD** exchange rate at 8:30 Tokyo Time and 17:30 Tokyo Time.
2. There might be some problems with those 17:30 data downloaded using the formula. The value we obtained in excel could not be found on the same day at Bloomberg Terminal, but could be found at 17:30 the following day. Every 17:30 value seemed to be placed on the following day. Therefore, we shifted all the 17:30 columns one row up in excel. $\Rightarrow$ This file is named as "CleanedCFETS".
3. We realized that the data of a few currencies were missing for 2016 and the beginning of 2017. Thus, we gave up 2016, but tried to fill up the beginning of 2017 for those particular currencies.
 - We did this by manually extracting data from Bloomberg Terminal.
 - If a 17:30 cell is empty, we manually check all the values for that date, and fill up the 17:30 value with the latest available value above it.
 - For example, the downloaded data looks like this, 2017-04-24 is empty

Date	USDSAR. 1	Bloomberg terminal
2017-04-19 00:00:00	3.7503	
2017-04-20 00:00:00	3.7503	
2017-04-21 00:00:00	3.7504	
2017-04-24 00:00:00		
2017-04-25 00:00:00	3.7504	
2017-04-26 00:00:00	3.7503	
2017-04-27 00:00:00	3.7503	

- Then we log onto Bloomberg Terminal, check BFIX page for this currency on 2017-04-21, 2017-04-22, 2017-04-23, 2017-04-24, 2017-04-25.
- We find that the value “3.7504” is the spot exchange rate for this currency on 2017-04-24 at 17:30.
- However, there is not any value for this currency on 2017-04-24 at 17:30 on the BFIX page.
- We look back in time, and find the latest available exchange rate for this currency appears on 2017-04-24 at 16:00.
- Thus, we can fill in the blank as the following, and take a note in the cell next to it.

A	R	S
Date	USDSAR. 1	Bloomberg terminal
2017-04-20 00:00:00	3.7503	
2017-04-21 00:00:00	3.7504	2017/4/21 16:00
2017-04-24 00:00:00	3.7504	2017/4/24 17:30
2017-04-25 00:00:00	3.7504	

- We cleaned 830 data for MYR, KRW, AED, SAR; 1730 data for AED, MYR, SAR.
- $\Rightarrow$ These new files are named as “830_concat(manual check).xlsx”, ”1730_concat(manual check).xlsx”
- Remove the comment column ”Bloomberg terminal” in the file to prepare next step calculation, we had “830_concat_check_nocomment.xlsx”, ”1730_concat_check_nocomment.xlsx”

3 Calculating X_t

- The weights applied to X_t could be retrieved from China Foreign Exchange Trade System
- The index-implied dollar index, defined by

$$X_t \equiv \left(\frac{S_t^{CP, EUR/CNY}}{S_t^{CP, USD/CNY}} \right)^{\frac{w_{EUR}}{1-w_{USD}}} \left(\frac{S_t^{CP, JPY/CNY}}{S_t^{CP, USD/CNY}} \right)^{\frac{w_{JPY}}{1-w_{USD}}} \dots$$

- Instead of using those “per CNY” values, we directly applied the “per USD” values retrieved from Bloomberg in Section 1.

-

$$\frac{S_t^{CP, EUR/CNY}}{S_t^{CP, USD/CNY}} = S_t^{CP, EUR/USD}$$

-

$$\begin{aligned} X_t &= \left(\frac{S_t^{EUR/CNY}}{S_t^{USD/CNY}} \right)^{\frac{w_{EUR}}{1-w_{USD}}} \left(\frac{S_t^{JPY/CNY}}{S_t^{USD/CNY}} \right)^{\frac{w_{JPY}}{1-w_{USD}}} \dots \\ &= \left(S_t^{EUR/USD} \right)^{\frac{w_{EUR}}{1-w_{USD}}} \left(S_t^{JPY/USD} \right)^{\frac{w_{JPY}}{1-w_{USD}}} \dots \end{aligned}$$

- Thus, there will be two X_t exist for one day and are calculated separately, morning $X_{t,830}$ (saved as "Xt830" in code), night $X_{t,1730}$ (saved as "Xt1730" in code).

3.1 Rescaling

- Whenever the Central Bank updated the CFETS basket, either adjusted weights of individual currencies or changed the listed currencies, we need to perform rescaling.
- For example, since the table of CFETS basket 2020 is different from the table of CFETS basket 2019, we should rescale the X_t values for 2020 by the following steps:
 1. Extract all the 8:30 exchange rate on *Dec 31st, 2019*
 2. Apply weight of 2020 to the 8:30 data *Dec 31st, 2019* $\Rightarrow$ a value of X (let's call it $X_{rescale}$)
 3. Previously, we have already obtained the $X_{Dec.31,830}$, i.e. Weight of 2019 applied to the last day of 2019 $\Rightarrow$ call it X_{actual}
 4. The *rescaling factor* is calculated as $\frac{X_{actual}}{X_{rescale}}$
 5. The whole column of X_t 2020 is multiplied by this *rescaling factor*
 6. Repeat the above process for $X_{t,1730}$ as well. Note: We calculate $X_{t,830}$ and $X_{t,1730}$ independently.
- In the end, we rescaled 2020, 2021, 2022, 2023 and 2024

3.2 Cleaning the 1730 X_t

- In some cases, namely New Year, Easter, Christmas, there are discrepancies between $X_{t,830}$ and $X_{t,1730}$ with respect to dates. I manually cleaned those by moving the incorrect $X_{t,1730}$ to the following day.
- For example, the current data frame looks like this

	Date	830_XtCFETS	1730_XtCFETS
74	2017-04-13	11.18465	11.16517
75	2017-04-14	NA	11.16743
76	2017-04-17	11.16799	NA

- This is another problem with the 1730 data we retrieved from Bloomberg.
 - In fact, the spot exchange rates were not available on 2017-04-14, but were available on 2017-04-17.
 - Although we cleaned the 1730 data as in Section 1 step2, in some cases around holidays there were problems with the date alignment as illustrated.
 - Therefore, we manually move the value 11.16743 to 2017-04-17.

	Date	830_XtCFETS	1730_XtCFETS
74	2017-04-13	11.18465	11.16517
75	2017-04-14	NA	NA
76	2017-04-17	11.16799	11.16743

- After the aligning the dates, remove all the NA values.

4 Pillar Construction

4.1 Cleaning close rate

- By definition, we used the CNY/USD spot rate at 17:30 Tokyo time as our close rate.
- We had to clean the dates for 17:30 data just as Section 2.2, and we logged onto Bloomberg terminal to confirm the dates.

A	B	
Date	close	Date
2017/4/3	6.8961	2017/3/31
2017/4/6	6.8999	2017/4/5
2017/4/7	6.8989	2017/4/6
2017/4/10	6.9056	2017/4/7
2017/4/11	6.9032	2017/4/10
2017/4/12	6.8925	2017/4/11
2017/4/13	6.8828	2017/4/12
2017/4/14	6.8872	2017/4/13
2017/4/18	6.886	2017/4/17
2017/4/19	6.8869	2017/4/18
2017/4/20	6.8839	2017/4/19
2017/4/21	6.883	2017/4/20
2017/4/24	6.8858	2017/4/21
2017/4/25	6.8861	2017/4/24
2017/4/26	6.8904	2017/4/25
2017/4/27	6.8922	2017/4/26
2017/4/28	6.8976	2017/4/27
2017/5/1	6.8967	2017/4/28
2017/5/3	6.892	2017/5/2
2017/5/4	6.8968	2017/5/3
2017/5/5	6.8999	2017/5/4

⇒

4.2 Combine data needed

- After rescaling $X_{t,830}$ and $X_{t,1730}$, cleaning $X_{t,1730}$ and cleaning close rate, we combined X_t , close rate, and central parity rate (CPR) into one data frame by dates.

4.3 Remove Chinese Public Holidays

- We observe that CPR values are not available during public holidays, therefore we remove the rows where CPR is NA.

Up to this stage, we can export the file “before pillar construction (2017-2024)”.

4.4 Construct pillars

- Note the dates may not be consistent since we removed Chinese Holidays.
 - For example, here over the Labor’s Day Holiday, we do not have data on 2017-04-29, 2017-04-30, 2017-05-01.
 - Thus, 2017-04-28 is treated as t and 2017-05-02 is treated as $t+1$ when we are constructing night pillar.

	Date	close	USD/CNY	830_XtCFETS	1730_XtCFETS
data_2017.83	2017-04-27	6.8922	6.8896	11.13190	11.13619
data_2017.84	2017-04-28	6.8976	6.8931	11.15265	11.15216
data_2017.86	2017-05-02	6.8967	6.8956	11.15227	11.13816
data_2017.87	2017-05-03	6.8920	6.8892	11.13094	11.14449

- S_t^{CP} refers to the PBOC’s central parity for RMB per dollar.
- S_t^{CL} : close rate from 4.1

Note: w_{USD} may differ for different years

The following is the notation used in pillar construction:

Data Name in Code	Original Formula
day change	$\left(\frac{X_{t,1730}}{X_{t,830}}\right)^{1-w_{USD}}$
night change	$\left(\frac{X_{t+1,830}}{X_{t,1730}}\right)^{1-w_{USD}}$
StDT	$S_t^{DT} = S_t^{CP} \left(\frac{X_{t,1730}}{X_{t,830}}\right)^{1-w_{USD}}$
St+1NT	$S_{t+1}^{NT} = S_t^{CP} \left(\frac{X_{t+1,830}}{X_{t,1730}}\right)^{1-w_{USD}}$
Sbart+1	$\bar{S}_{t+1} = \frac{S_{t+1}^{NT} S_t^{DT}}{S_t^{CP}}$

Up to this stage, we can export the file ”new_pillar1720.xlsx” and ”new_pillar2124.xlsx”, and combining these two files we have ”new_pillar1724.xlsx”

5 Rolling Window Regression & Prediction

5.1 Regimes Regression

Regime 1

$$\log \left(\frac{S_{t+1}^{CP}}{S_t^{CP}} \right) = \alpha \log \left(\frac{S_t^{CL}}{S_t^{CP}} \right) + \beta \log \left(\frac{\bar{S}_{t+1}}{S_t^{CP}} \right) + \epsilon_t$$

Regime 2

$$\log \left(\frac{S_{t+1}^{CP}}{S_t^{CP}} \right) = \alpha \log \left(\frac{S_t^{CL}}{S_t^{CP}} \right) + \beta_{NT} \log \left(\frac{S_{t+1}^{NT}}{S_t^{CP}} \right) + \epsilon_t$$

Regime 3

$$\log \left(\frac{S_{t+1}^{CP}}{S_t^{CP}} \right) = \alpha \log \left(\frac{S_t^{CL}}{S_t^{CP}} \right) + \beta_{NT} \log \left(\frac{S_{t+1}^{NT}}{S_t^{CP}} \right) + \beta_{DT} \log \left(\frac{S_t^{DT}}{S_t^{CP}} \right) + \epsilon_t$$

For the above regression, take 30 days as a window to select the variables and perform rolling-window regression.

For each rolling-window regression, 30 rows are included, $t=i-n$, $n=0,1,2,\dots,29$, and the regression result would be aligned with row i . Note that all i and t here mean row number in the data, not continuous date.

5.2 Regression Data

Data Name in Code	Original Formula
CP _{t+1} /CP	$\log \left(\frac{S_{t+1}^{CP}}{S_t^{CP}} \right)$
CL/CP	$\log \left(\frac{S_t^{CL}}{S_t^{CP}} \right)$
NT _{t+1} /CP	$\log \left(\frac{S_{t+1}^{NT}}{S_t^{CP}} \right)$
DT/CP	$\log \left(\frac{S_t^{DT}}{S_t^{CP}} \right)$

5.3 Baseline Regression results

Whole sample regression with Regime 1.

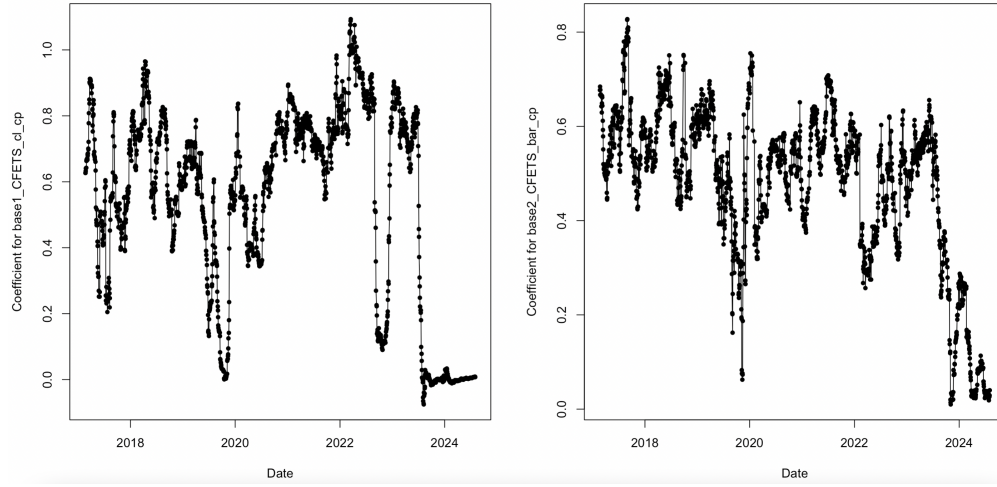
1. If we apply data range 2017-2020, the regression output is:

	<i>Dependent variable:</i>
	CP_logdiff_CFETS
base1_CFETS_cl_cp	0.435*** (0.013)
base2_CFETS_bar_cp	0.579*** (0.017)
Observations	957
R ²	0.757
Adjusted R ²	0.757
Residual Std. Error	0.001 (df = 955)
F Statistic	1,491.523*** (df = 2; 955)
<i>Note:</i>	* p<0.1; ** p<0.05; *** p<0.01

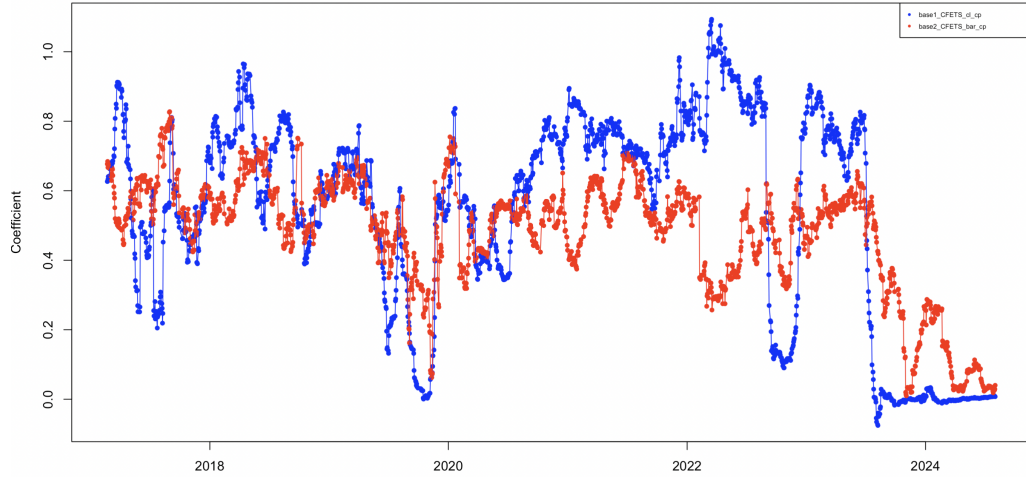
2. If we apply data range 2017-2024, the regression output is

	<i>Dependent variable:</i>
	CP_logdiff_CFETS
base1_CFETS_cl_cp	0.100*** (0.007)
base2_CFETS_bar_cp	0.600*** (0.016)
Observations	1,795
R ²	0.487
Adjusted R ²	0.487
Residual Std. Error	0.002 (df = 1793)
F Statistic	852.125*** (df = 2; 1793)
<i>Note:</i>	* p<0.1; ** p<0.05; *** p<0.01

3. The Adjusted R^2 fell a lot once we incorporate data after 2020, we wanted to see where the problem started, thus we ran 30-day rolling window regressions with Regime 1 and made the flowing plots
- The value of the two coefficients plotted separately



- The value of the two coefficients plotted together



5.4 Subperiod regressions 2017-2020

- JWY (2022) has two sets of regressions with sample period covering 2015-12-11 to 2020-12-31.

1. **Baseline Regression** for the whole sample:

$$\log \left(\frac{S_{t+1}^{CP}}{S_t^{CP}} \right) = \alpha \log \left(\frac{S_t^{CL}}{S_t^{CP}} \right) + \beta \log \left(\frac{\bar{S}_{t+1}}{S_t^{CP}} \right) + \epsilon_t$$

2. **Subperiod regression:**

- (a) Subperiod I (Regime 1): Dec 11, 2015 to Feb 19, 2017
- (b) Subperiod II (Regime 2): Feb 20, 2017 to May 25, 2017
- (c) Subperiod III (Regime 3): May 26, 2017 to Jan 8, 2018

- (d) Subperiod IV (Regime 2): Jan 9, 2018 to Aug 23, 2018
- (e) Subperiod V (Regime 3): Aug 24, 2018 to Oct 26, 2020
- (f) Subperiod VI (Regime 2): Oct 27, 2020 to Dec 31, 2020

5.5 Subperiod regressions 2021-2024

We had to determine which regime to use by ourselves for data from 2021 onwards.

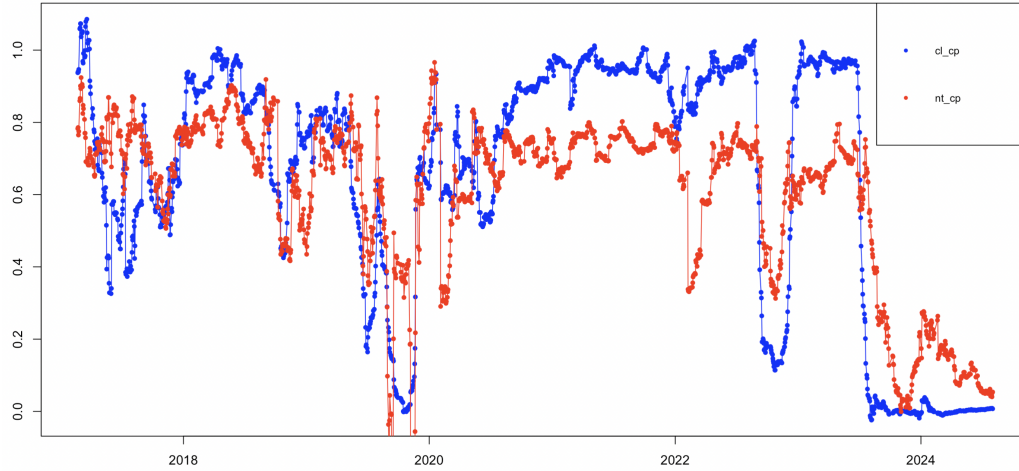
We applied rolling window regressions with window size of 30, and compare the adjusted R^2 under regime 2 and 3.

The procedure is as follows:

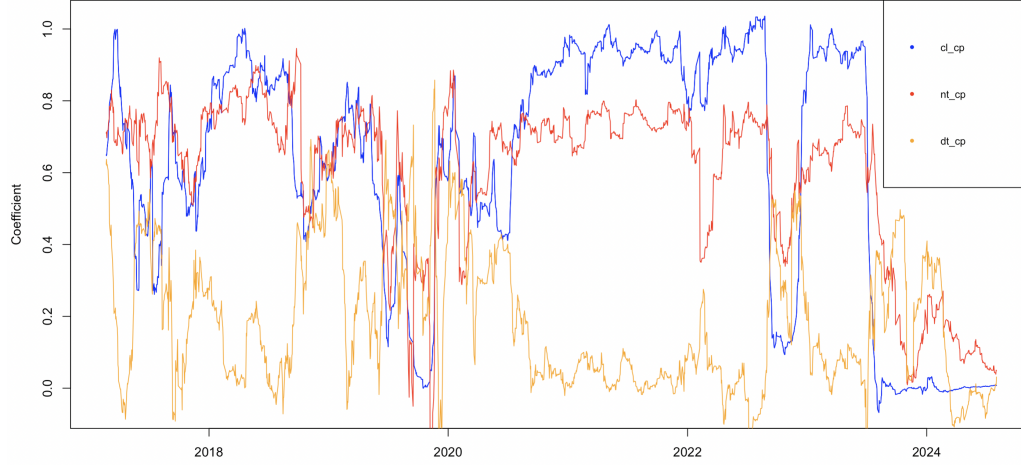
1. Extract a 30 day period, from day1 to day 30, apply regime 2, extract the adjusted R^2 of the period.
2. Roll to the next 30 day period, from day 2 to day 31, apply regime 2, extract the adjusted R^2 of the period.
3. This goes on and on until the last day.
4. Redo the above steps for regime 3.

We can generate the following plots

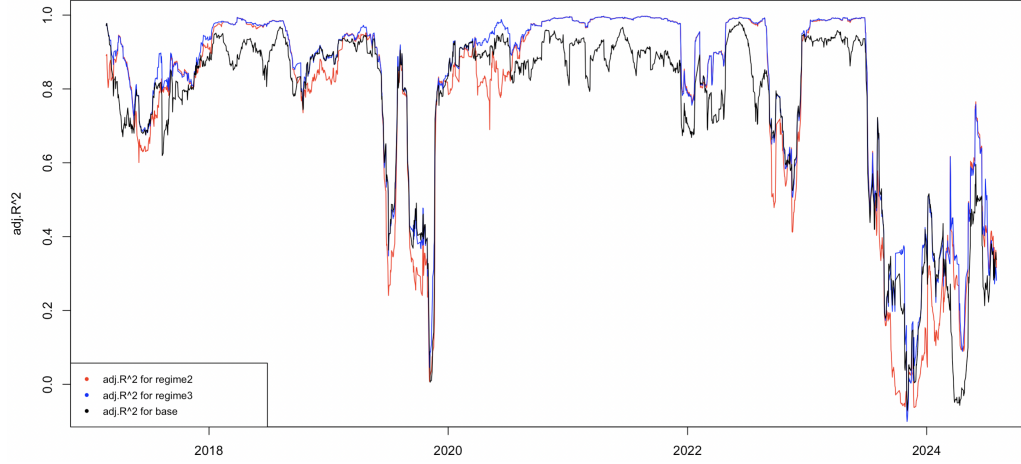
- Rolling window coefficients for Regime 2



- Rolling window coefficients for Regime 3



- Comparing the adjusted R^2 under the three regimes



5.6 Dividing Subperiod

- Although it seems that regime 3 outperforms regime 2 in most cases, our rule of thumb is to **utilize as few variables as we can** thus we try to use regime 2 as long as the adjusted R^2 for the period is above 0.86.
- Our result suggests that we should split the subperiods as:
 - (g) Subperiod VI (Regime 2): 2020-10-27 to 2021-12-08 (we extended the subperiod from the end of 2020 to the end of 2021)
 - (h) Subperiod VII (Regime 3: 2021-12-09 to 2022-03-17
 - (i) Subperiod VII (Regime 2): 2022-03-18 to 2022-08-31
 - (j) Subperiod IX (Regime 3): 2022-09-01 to 2022-12-4
 - (k) Subperiod X (Regime 2): 2022-12-05 to 2023-06-29

Here is a summary of all subperiods and corresponding regimes

Time Segment	Regime
2017-02-20 to 2017-05-25	Regime 2
2017-05-26 to 2018-01-08	Regime 3
2018-01-09 to 2018-08-23	Regime 2
2018-08-24 to 2020-10-26	Regime 3
2020-10-27 to 2021-12-08	Regime 2
2021-12-09 to 2022-03-27	Regime 3
2022-03-18 to 2022-08-31	Regime 2
2022-09-01 to 2022-12-04	Regime 3
2022-12-05 to 2023-06-28	Regime 2

5.7 Subperiod Regression Results

5.7.1 Regression table

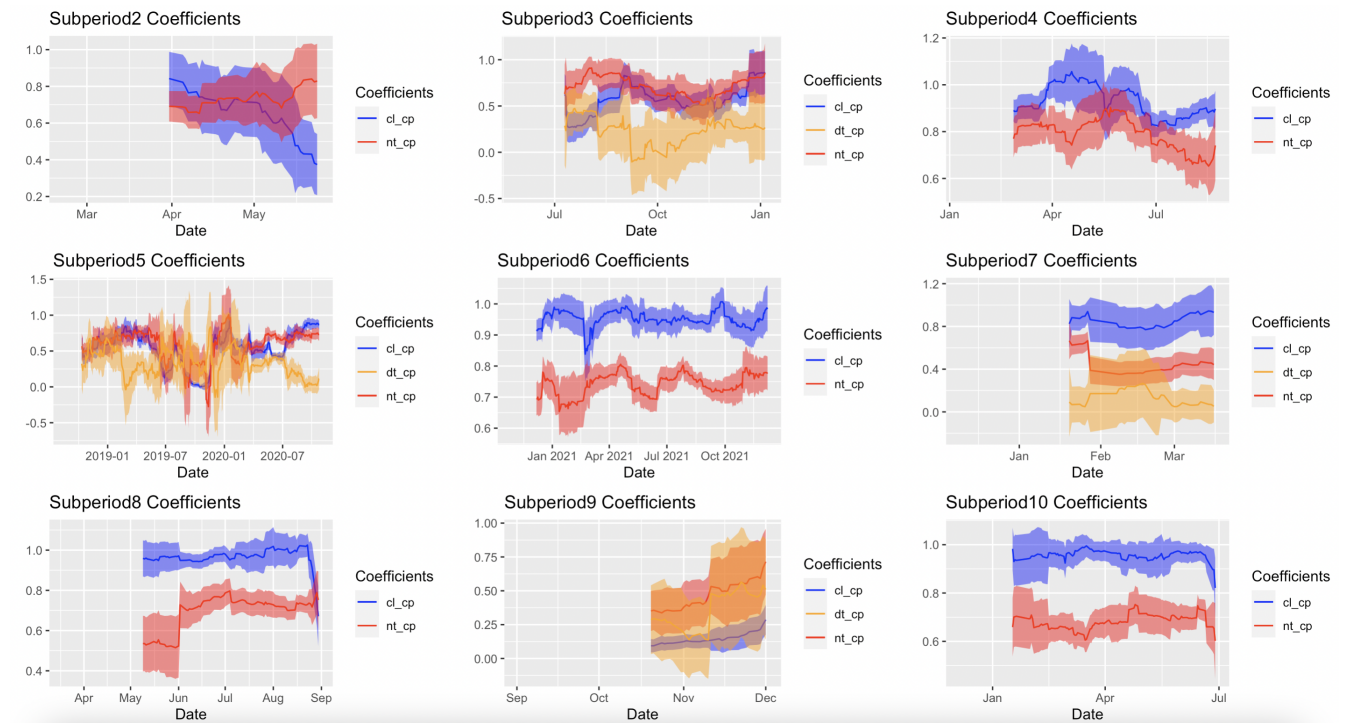
Dependent variable:									
	(1)	(2)	(3)	(4)	CP_logdiff (5)	(6)	(7)	(8)	(9)
cl_cp	0.464*** (0.060)	0.602*** (0.035)	0.886*** (0.014)	0.348*** (0.016)	0.945*** (0.007)	0.883*** (0.070)	0.912*** (0.023)	0.161*** (0.032)	0.945*** (0.020)
nt_cp	0.740*** (0.053)	0.724*** (0.046)	0.790*** (0.022)	0.607*** (0.029)	0.740*** (0.008)	0.505*** (0.049)	0.681*** (0.032)	0.555*** (0.082)	0.654*** (0.024)
dt_cp		0.225*** (0.073)		0.474*** (0.036)		0.060 (0.058)		0.477*** (0.131)	
Observations	64	152	151	519	271	64	110	60	133
R ²	0.785	0.808	0.971	0.719	0.988	0.835	0.957	0.646	0.960
Adjusted R ²	0.778	0.804	0.971	0.718	0.988	0.827	0.957	0.627	0.960
Residual Std. Error	0.001 (df = 62)	0.001 (df = 149)	0.0005 (df = 149)	0.001 (df = 516)	0.0002 (df = 269)	0.001 (df = 61)	0.001 (df = 108)	0.002 (df = 57)	0.001 (df = 131)
F Statistic	113.400*** (df = 2; 62)	208.592*** (df = 3; 149)	2,501.170*** (df = 2; 149)	440.519*** (df = 3; 516)	11,283.990*** (df = 2; 269)	103.195*** (df = 3; 61)	1,215.892*** (df = 2; 108)	34.674*** (df = 3; 57)	1,591.131*** (df = 2; 131)

Note:

*p<0.1; **p<0.05; ***p<0.01

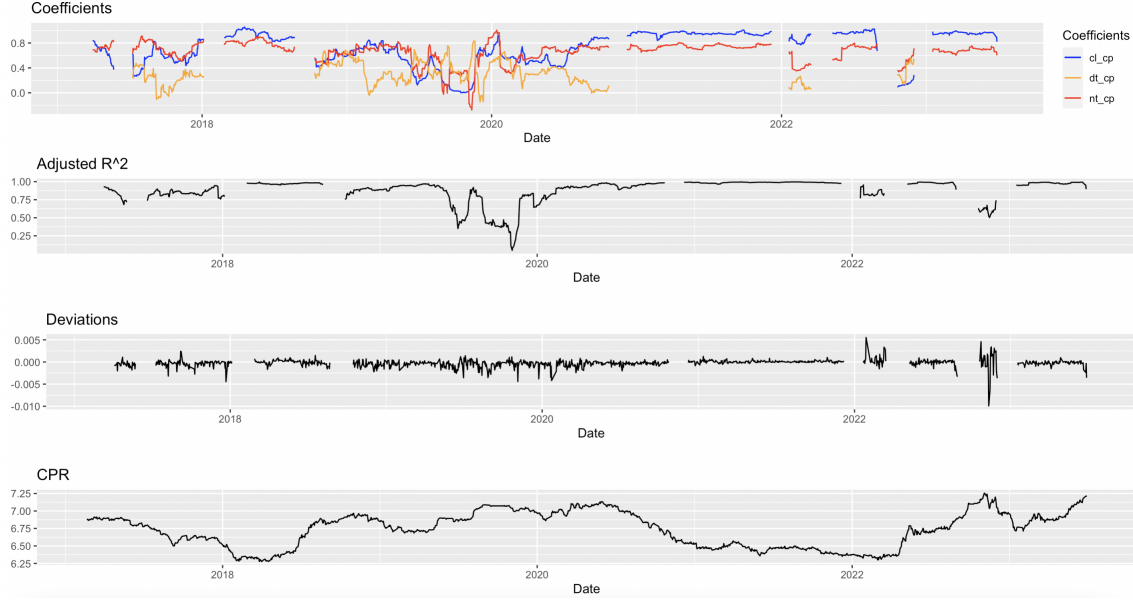
- Model (1) corresponds to (b)
- Model (2) corresponds to (c)
- Model (3) corresponds to (d)
- Model (4) corresponds to (e)
- Model (5) corresponds to (g)
- Model (6) corresponds to (h)
- Model (7) corresponds to (i)
- Model (8) corresponds to (j)
- Model (9) corresponds to (k)

5.7.2 Coefficient plots



- Plotted with 95% confidence intervals

5.7.3 Others



6 Rolling Window Forecast

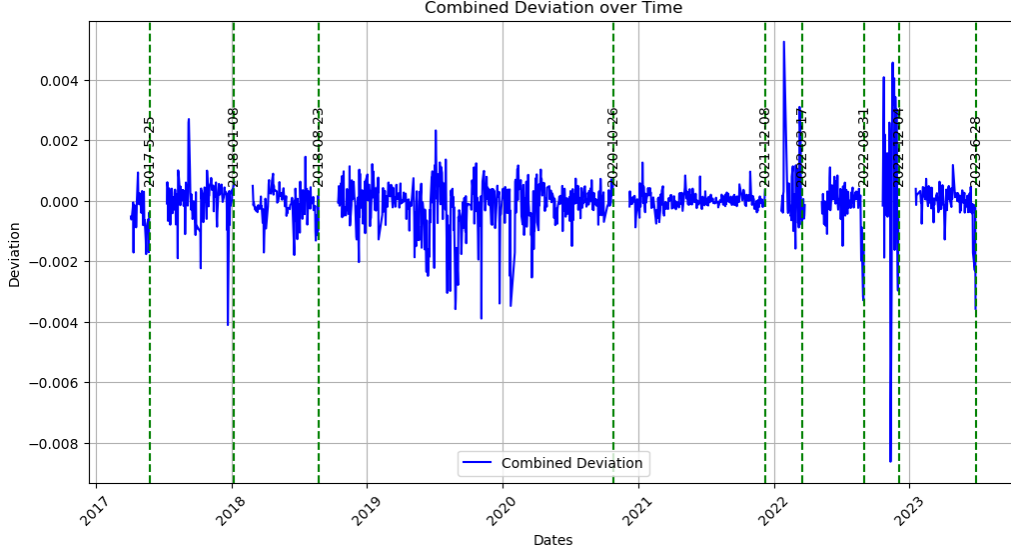
- We performed rolling window forecast within each subperiod.
- With a window size of 30, we run regression with the assigned regime of the subperiod, use the model generated by the 30 day period to predict $\log\left(\frac{S_{t+1}^{CP}}{S_t^{CP}}\right)$ of day 31.
- The procedure is as follows:
 - Extract the first 30 days of the subperiod, run regression with the sample by its regime.
 - Apply the generated regression output to the values of day 31, therefore we get the prediction of the dependent variable $\log\left(\frac{S_{t+1}^{CP}}{S_t^{CP}}\right)$ of day 31.
 - This process goes on and on until the last day in the subperiod.
 - The above steps applied to all the subperiods.

6.1 Deviation of Prediction

Calculate the deviation between the actual value $\log\left(\frac{S_{t+1}^{CP}}{S_t^{CP}}\right)$ and the prediction of the dependent variable $\log\left(\frac{S_{t+1}^{CP}}{S_t^{CP}}\right)$ given by the above rolling-window regression with corresponding regime regression, that is, the deviation for each rolling window regression i for the corresponding regime

regression:

$$\Delta_t = \log \left(\frac{S_{t+1}^{CP}}{S_t^{CP}} \right) - \log \left(\widehat{\frac{S_{t+1}^{CP}}{S_t^{CP}}} \right), t = i$$



7 CPR Deviation

7.1 Calculation of Log Deviation

Note: Since the first Δ_t appears on day 31 of each subperiod, the first $\widehat{S}_t^{CP}$ available is on day 32 of each subperiod.

Data Name in Code	Original Formula
CPR_estimate	$\widehat{S}_t^{CP} = e^{(\log(S_t^{CP}) - \Delta_{t-1})}$
CPR_deviation	$\text{CPR_deviation}_t = \log \left(\frac{S_t^{CP}}{\widehat{S}_t^{CP}} \right)$

8 Predictive Regression

8.1 Regression

For the following regression, we only consider the data starting from 2017-01-01, ending at 2020-12-31.

For $j=0,1,2,\dots,119$

$$\begin{aligned} \log \left(\frac{S_{t+j}^{CL}}{S_{t-1}^{CL}} \right) &= \alpha + \beta_1 \log \left(\frac{S_t^{CP}}{\widehat{S_t^{CP}}} \right) + \beta_2 \log(S_{t-1}^{CL}) + \beta_3 \log \left(\frac{S_{t-1}^{CL}}{S_{t-2}^{CL}} \right) + \beta_4 \log \left(\frac{S_t^{NT}}{\frac{S_{t-1}^{CP}}{S_{t-1}^{NT}} \frac{S_{t-1}^{NT}}{S_{t-2}^{CP}}} \right) + \epsilon_t \\ &= \alpha + \beta_1 \text{CPR_deviation}_t + \beta_2 \log(S_{t-1}^{CL}) + \beta_3 \text{CLpillar_change}_t + \beta_4 \text{NTpillar_change}_t + \epsilon_t \end{aligned}$$

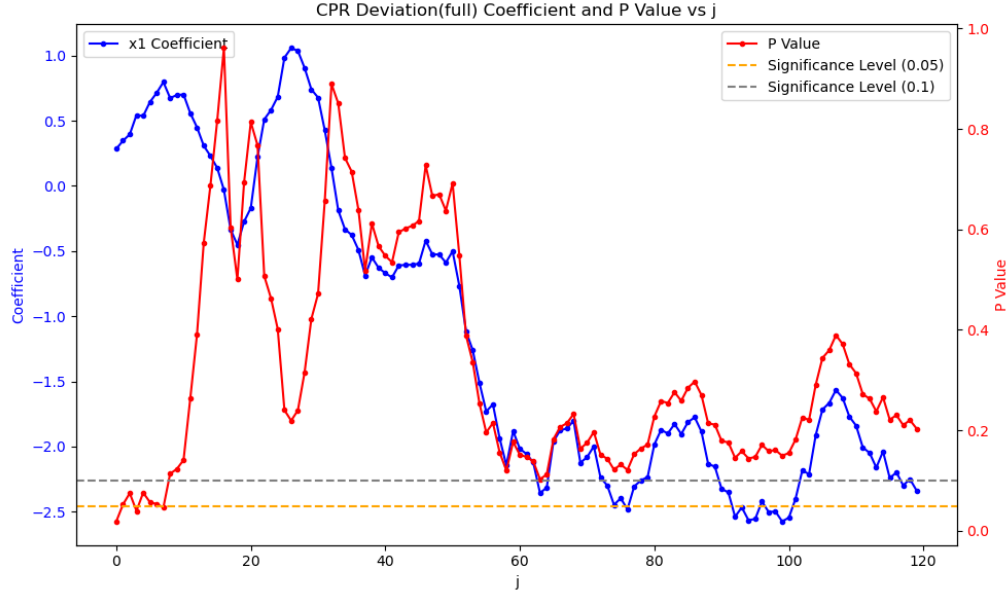
Now we have 120 regression results and for each $j(j=0,1,2,\dots,119)$ we have a corresponding result β_1 and p-value of β_1 . Then we can draw a graph with x-axis is j and y-axis is p-value of β_1 .

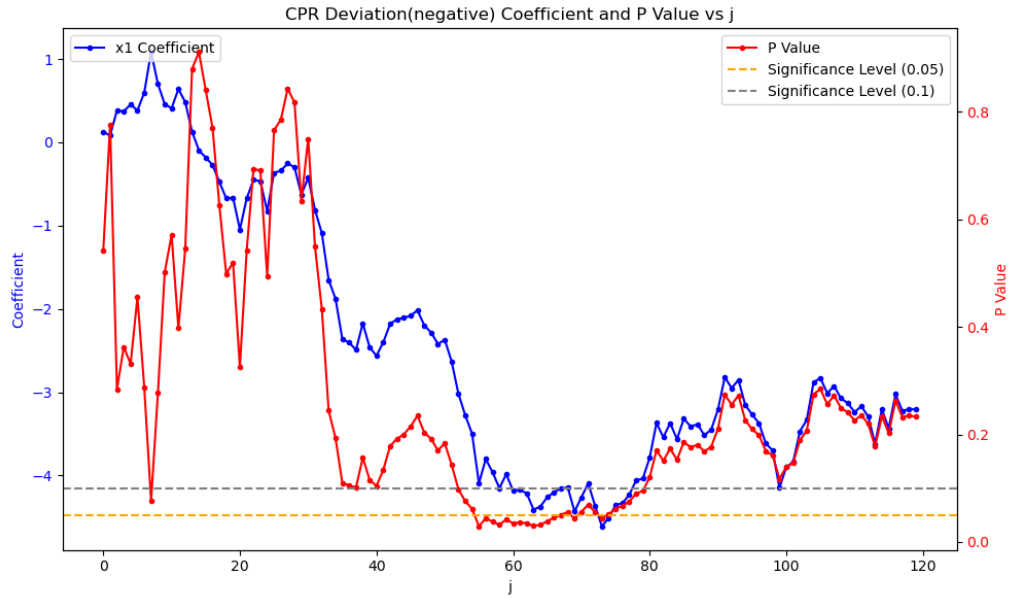
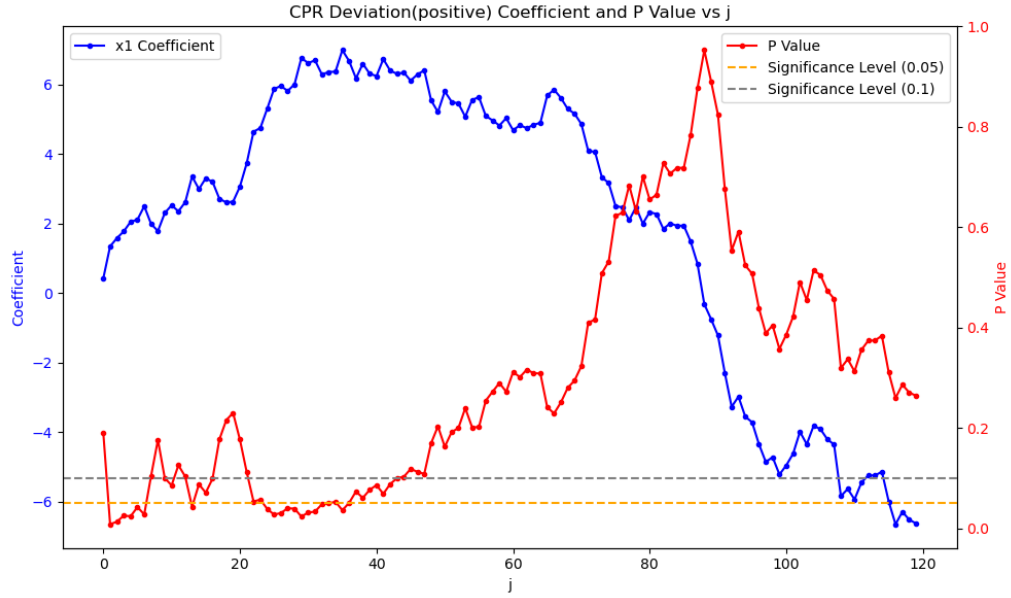
To further study the regression result for all data that CPR_deviation is positive or negative. We use the dataframe used in the previous regression and then do one more step to selecting the sign of $\log \left(\frac{S_t^{CP}}{\widehat{S_t^{CP}}} \right)$:

For positive case, we only keep the set of data with positive $\log \left(\frac{S_t^{CP}}{\widehat{S_t^{CP}}} \right)$ used in the previous regression and then to do the same regression.

For negative case, we only keep the set of data with negative $\log \left(\frac{S_t^{CP}}{\widehat{S_t^{CP}}} \right)$ to do the same regression. The rest of the steps are same as before.

In the end, we study the regression with full, positive and negative $\log \left(\frac{S_t^{CP}}{\widehat{S_t^{CP}}} \right)$ these three ways and get three graphs.





8.2 Regression Data

Data Name in Code	Original Formula
CPR_deviation	$\text{CPR_deviation}_t = \log \left(\frac{S_t^{CP}}{\overline{S_t^{CP}}} \right)$

Data Name in Code	Original Formula
CLt-1	$\log(S_{t-1}^{CL})$
CLpillar_change	$\text{CLpillar_change}_t = \log\left(\frac{S_{t-1}^{CL}}{S_{t-2}^{CL}}\right)$
NTpillar_change	$\text{NTpillar_change}_t = \log\left(\frac{S_t^{NT}}{S_{t-1}^{CP}}\right) - \log\left(\frac{S_{t-1}^{NT}}{S_{t-2}^{CP}}\right) = \log\left(\frac{\frac{S_t^{NT}}{S_{t-1}^{CP}}}{\frac{S_{t-1}^{NT}}{S_{t-2}^{CP}}}\right)$

9 HKMA Liquidity Regression

9.1 Regression from paper

$$z_{t+h} = \alpha_h + \beta_h \Delta e_t + \gamma_h e_{t-1} + \delta_h z_{t-1} + \text{controls}_{t-1} + \text{error}_t^h$$

9.2 Regression (we run)

For $H = 0, 1, 2, 3, 4, 5$

$$\begin{aligned}
Z_{t+H} &= \alpha + \beta_H \log\left(\frac{S_t^{CP}}{S_t^{CP}}\right) + \beta_2 Z_{t-1} + \beta_3 \text{Drawings}_{t-1} + \beta_4 \text{HIBOR3M}_{t-1} + \beta_5 \text{SHIBOR}_{t-1} \\
&\quad + \beta_6 \log(S_{t-1}^{CL}) + \beta_7 \log\left(\frac{S_{t-1}^{CL}}{S_{t-2}^{CL}}\right) + \beta_8 \log\left(\frac{\frac{S_t^{NT}}{S_{t-1}^{CP}}}{\frac{S_{t-1}^{NT}}{S_{t-2}^{CP}}}\right) + \epsilon_t \\
&= \alpha + \beta_1 \text{CPR_deviation}_t + \beta_2 Z_{t-1} + \beta_3 \text{Drawings}_{t-1} + \beta_4 \text{HIBOR3M}_{t-1} + \beta_5 \text{SHIBOR}_{t-1} \\
&\quad + \beta_6 \log(S_{t-1}^{CL}) + \beta_7 \text{CLpillar_change}_t + \beta_8 \text{NTpillar_change}_t + \epsilon_t
\end{aligned}$$

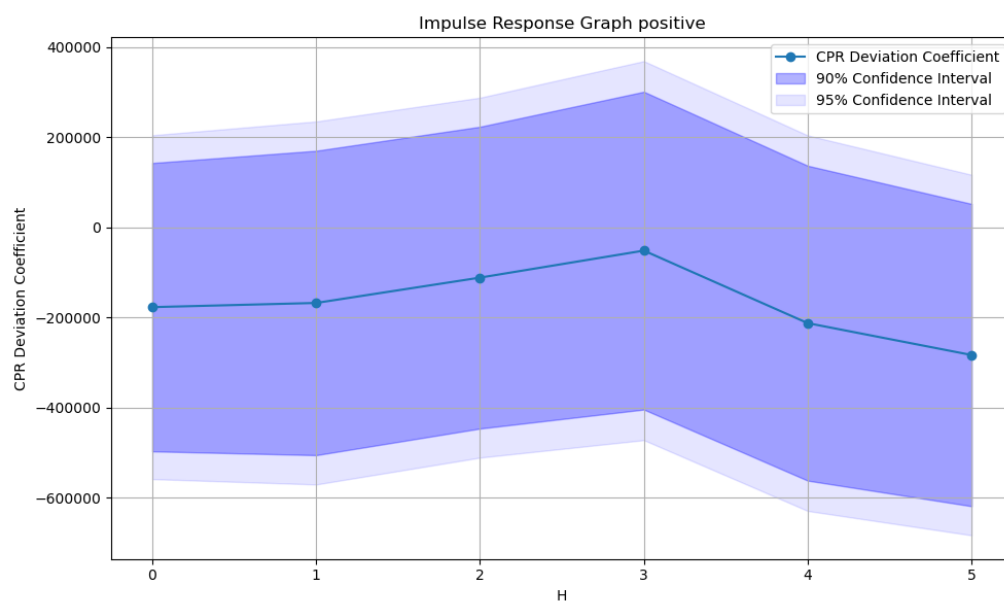
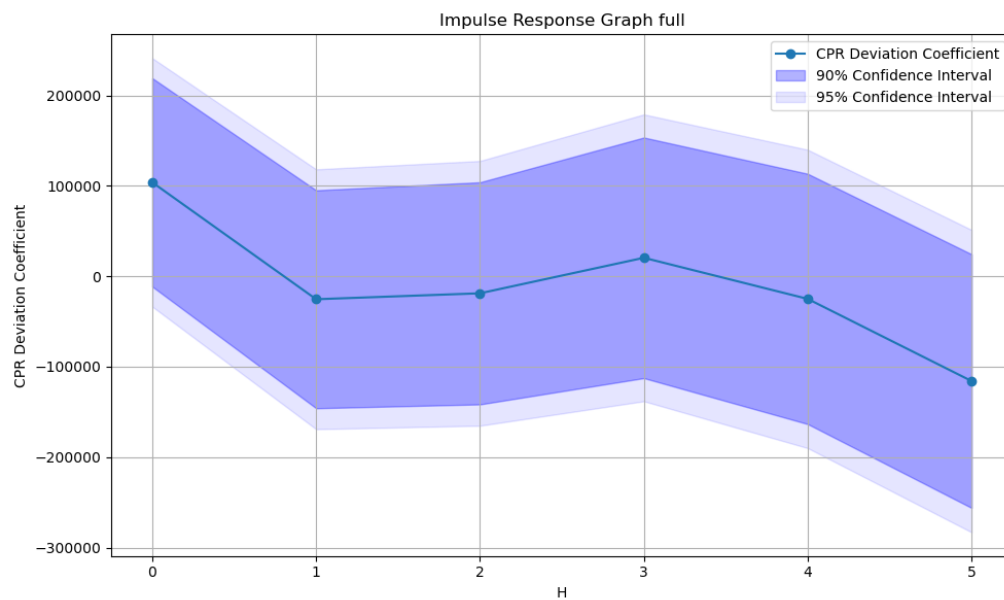
Now we have 6 regression results and for each H ($H = 0, 1, 2, 3, 4, 5$) we have a corresponding result β_H and its 90% confident interval and 95% confident interval. Then we can draw a impulse response graph with x-axis is H and y-axis is coefficient of β_H with confident interval 90% and 95%.

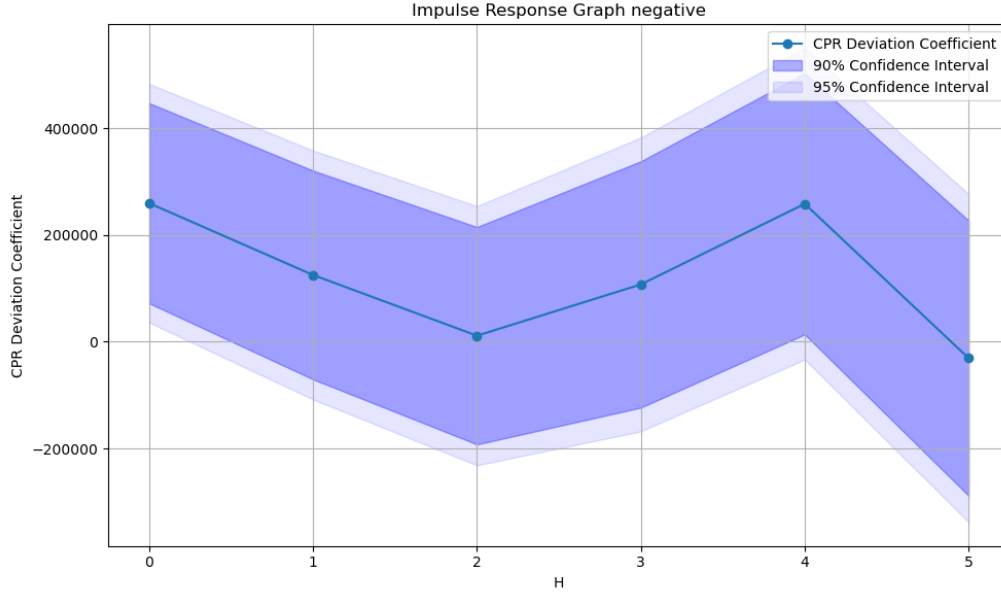
To further study the regression result for all data that CPR_deviation is positive or negative. We still use the dataframe used in the previous regression and then do one more step to selecting the sign of $\log\left(\frac{S_t^{CP}}{S_t^{CP}}\right)$:

For positive case, we only keep the set of data with positive $\log\left(\frac{S_t^{CP}}{S_t^{CP}}\right)$ to do the same regression.

For negative case, we only keep the set of data with negative $\log\left(\frac{S_t^{CP}}{S_t^{CP}}\right)$ to do the same regression. The rest of the steps are same as before.

In the end, we study the regression with full, positive and negative $\log\left(\frac{S_t^{CP}}{S_t^{CP}}\right)$ these three ways and get three impulse response graphs.





9.3 Raw Data Downloading and Calculation

By using HKMA API.ipynb, we are able to download "rmb_liquidity_statistics.xlsx" from <https://api.hkma.gov.hk/public/market-data-and-statistics/daily-monetary-statistics/usage-rmb-liquidity-fac>

9.3.1 Z_t : Liquidity

Z_t is the corresponding data in column "plp_fac_at_1600" in the downloaded file.

9.3.2 Drawings_t: Drawings from the HKMA's discount window facility

For each t , in the downloaded file, we use a_t , b_t , c_t , d_t to represent the data in column "intraday_repo_at_0900", "intraday_repo_at_1100", "intraday_repo_at_1400", "intraday_repo_at_1600"

Then, we can have

$$g_t = \max\{a_t, b_t, c_t, d_t\}$$

For Drawings,

$$\text{Drawings}_t = \frac{1}{10} \sum_{k=1}^{10} g_{t-k}$$

These steps are included in "drawings from the HKMA's discount window facility.ipynb" and the output data is in the last column of the output file "drawings from the HKMA's discount window facility.xlsx"

9.3.3 HIBOR3M_t: HIBOR as 3 month CNH interbank rates

This is downloaded from the database terminal and saved as "HIBOR3M NAasNA.xlsx"

9.3.4 SHIBOR_t: SHIBOR as 3 month CNY interbank rates

This is downloaded from <https://www.shibor.org/shibor/dataservicesen/>
saved as "Shibor_Historical_Data_2017_2020.xlsx"

9.3.5 Other Control Variables

Other control variables includes CL, CLpillar_change and NTPillar_change used in section 8.2 before.

9.4 Clean data for regression

Use "Merging Regression Data.ipynb" to take the union of dates of all independent variables data for the regression.

Combine the above into one data frame by dates and get output "regression_data.xlsx"

9.4.1 The controls have to be placed at t-1

The following step and the regression is included in "Regression HKMA dont fill NA.ipynb"

Only the CLt-1 has already been placed on t-1, but others are on t. In the regression formula, the controls should be placed on t-1. We do the following steps:

- Take lag for controls and get "Drawings_t_minus_1", "HIBOR3M_t_minus_1", "SHIBOR_t_minus_1", "CLpillar_change_t_minus_1", "NTPillar_change_t_minus_1" in code.
- The dependent variables are liquidity Z_{t+H} where $H = 0, 1, 2, 3, 4, 5$. Thus we take lead for liquidity, with lead = 1, 2, 3, 4, 5.
- At the same time, Z_{t-1} is one of the controls on the right. Thus take lag=1 for liquidity here as well and get "Liquidity_t_minus_1" in code.